

ULTRALEARNING

Master Hard Skills, Outsmart the Competition, and Accelerate Your Career

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The Big “So What”

Can you get an MIT-level education without attending MIT, become fluent in a new language in 3 months, or develop a multi-million video game by yourself? The answer is “yes”. In this book, Young shares the strategies and tactics used by ultralearners to master hard skills more quickly, accelerate their careers, become more productive and fulfilled.



Introduction



Ultralearning is a strategy for acquiring skills and knowledge in a self-directed, intense way. It’s about deciding what you want to learn, then committing to a period of deep, focused learning activities that push you to your limits. That’s the best way to get in flow and compress your learning into the shortest possible time.

- Young was interested in computer science, but couldn’t afford to attend MIT. So, he embarked on a self-education project, researching free MIT classes online, focusing on the final exams and working backward to figure out what he had to learn and how. In just 1 year, he could pass the final exams for the MIT computer science programme. He went on to learn 4 languages in 12 months, and taught himself to draw portraits in 30 days.
- Along the way, he met and researched other “ultralearners”, including polyglots who speak many languages, mathematical geniuses, game show winners, etc. Based on his observations and experience, Young identified the 9 common principles in this book.

KEY QUOTES

“If you can master the personal tools to learn new skills quickly and effectively, you can compete more successfully in this new environment.”

“With ultralearning, deeply and effectively learning things is always the main priority.”

“Learning well isn’t just about following a set of prescriptions. You need to try things out for yourself, think hard about the nature of the learning challenges you face, and test solutions to overcome them.”



Ultralearning is not easy, but it can be extremely rewarding.

- Ultralearning is possibly one of the best strategies to adapt and compete in a dynamic world. With the rapid pace of change, those who can't renew themselves quickly get left behind. Ultralearning helps you to learn more in a fraction of the time, so you can advance your career quickly or even transit to a new career.
- You can also use it to master any area of personal interest—be photography or culinary skills—to feel more confident and fulfilled.

Every learning project is different, and you won't learn effectively by rigidly following a fixed sequence of steps. However, you can apply the 9 ultralearning principles below to learn optimally in any area. The key is to decide what matters to you, and take responsibility to achieve those goals.

KEY QUOTES

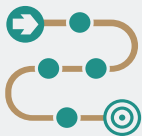
"The best ultralearners are those who blend the practical reasons for learning a skill with an inspiration that comes from something that excites them."



The 9 Ultralearning Principles

THE 9 ULTRALEARNING PRINCIPLES

Metalearning



Start with a Road Map

Focus



Sharpen your Ability to Concentrate

Directness



Go Straight into Application

Drill



Attack your Weakest Point

Retrieval



Use Self-Testing to Learn

Feedback



Don't Dodge the Punches

Retention



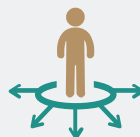
Stop the Leakages

Intuition



Dig Deep, then Build Up

Experimentation



Explore Beyond your Comfort Zone



1. Metalearning: Start with a Road Map



The first step in any ultralearning project is to figure out the most effective and efficient way to learn that subject.

- Formal education often uses a one-size-fits-all approach to cater to a range of students. However, you can customize your self-learning project to fit your specific needs and abilities. By investing some time upfront to develop a learning map, you will (i) learn much faster for your current project, and (ii) improve your long-term learning ability to learn faster in any area.



Develop your metalearning map with 3 specific questions:

- Define why you want to learn that subject. Your goal or desired outcomes will determine the type of skills and knowledge you need. Try *interviewing experts* in the field to refine your learning plan. For example, you can speak with a successful architect to confirm if learning design skills will indeed help you to become a good architect. Don't be afraid to reach out to experts via social media, email or other channels: many of them will be willing to do a 15 minute phone/video call to help you.
- Once know your "why", you can figure out what you must learn.
 - (i) On a piece of paper, create 3 columns labeled: *concepts, facts, and procedures*.
 - *Concepts*: What are the ideas you must understand? For example, to learn math you need to grasp concepts like equality and symbolic representation.
 - *Facts*: What are the things that you must memorize? For example, languages involve vocabulary, pronunciation and grammar.
 - *Procedures*: What are the actions that must be practiced?
 - (ii) *Identify your bottlenecks*, i.e. the concepts, facts and procedures that are the hardest for you. Research the

KEY QUOTES

"Metalearning...forms the map, showing you how to get to your destination without getting lost."

"Each project you complete will give you new tools to tackle the next, starting a virtuous cycle."

tools/methods to help you tackle those bottlenecks, so you have a more realistic learning map.

- To figure out how to learn what's required:
 - (i) *Benchmark* how people typically learn that subject. For example, you can research how it's usually taught in schools or universities, examine the topics in textbooks, or course syllabuses found on websites.
 - (ii) Modify the default curriculum, by *emphasizing or excluding* certain topics to suit your personal needs/preferences. For example, if you're learning French because you're going to Paris in 2 weeks, you'd want to focus more on pronunciation than spelling.



You do *not* need a perfect learning map. The idea is to make sure you've evaluated your options instead of jumping into whatever's easy or convenient. You can always add more details as you go along. So, is there an optimal amount of planning?

- As a rule of thumb, invest about 10% of your expected learning time into the initial research and planning.
- Then, regularly review your progress to see if diminishing returns have set in. If so, pause to research new tools/techniques.



2. Focus: Sharpen your Ability to Concentrate



Focused learning delivers superior results. You can improve your ability to focus through practice—by dedicating chunks of time to focus fully on what you're trying to learn.



Ultralearners overcome 3 key obstacles to concentration:

- Procrastination: When it's time to start on your ultralearning project, you may find yourself putting it off, e.g. you tell yourself that you need a break, you suddenly feel hungry, or feel an urge to pack the table. We often procrastinate without realizing it, so the first step is to recognize when you're procrastinating. Then, you can overcome it with the following tactics:

KEY QUOTES

"Find the sweet spot between insufficient research and analysis paralysis."

"Most motives to procrastinate are silly when you verbalize them, yet that doesn't stop them from ruling your life."

- (i) Acknowledge the unpleasant aspects of the task (e.g. you worry that you may not do a good job, or feel uncertain about where to start). Tell yourself to just bear with it for *5-10 minutes*. Once you get started, the unpleasant feelings usually fade away and you end up doing much more.
- (ii) To stay focused, promise yourself a 5-minute break each time you complete 25 minutes on the project.
- (iii) Set a rule that you can only quit after you've *achieved a milestone*, e.g. memorized a vital piece of information.
- *Distractions*: Most learning tasks can be finished in 50-60 minutes of focused learning. Unfortunately, we get distracted easily, e.g. we find ourselves daydreaming or checking our phone. Remove these 3 common sources of distractions:
 - (i) Set up your ideal learning *environment*, e.g. have your writing materials at hand, and turn off distractions like the TV/phone.
 - (ii) Choose the *learning tool* that best grabs your attention. For example, do you focus better when you read, listen to an audio or watch a video? You can also make notes as you read to help you to focus.
 - (iii) *Negative emotions* (e.g. anger, frustration) can hijack your mind. Acknowledge the feeling, allow it to pass, and gently refocus back on your project.
- *Variety*: Often, simpler tasks (e.g. throwing darts or shooting basketballs) are best performed in a stimulating environment (e.g. upbeat music that gives you a burst of mental arousal). Complex tasks (e.g. writing an essay or solving a complex problem) may require relaxed focus and a quieter environment. Experiment to find the ideal states of mental arousal you need for different learning activities.

KEY QUOTES

"Remember: even if your learning is intense, your use of the skill later on will not be."

"A clear, calm mind is best for focusing on almost all learning problems."



3. Directness: Go Straight into Application



Consider the situation/context where you want to apply your learning. Then, plunge right in and acquire hands-on experience by learning to do the tasks in a similar situation. Don't settle for other convenient or comfortable activities.

- For example, a fresh graduate couldn't get a job as an architect. He realized that many architecture firms were using a complex design software, so he learned the software and used it to design his own building. With this portfolio, he landed a job quickly.
- The fastest way to learn something is by doing it directly. Yet, most schools use an indirect approach: they teach facts, concepts and skills in a classroom setting, and assume students can transfer the learning to the real world. In reality, different situations come with nuances that require different solutions, and it's not easy to apply theoretical knowledge to a specific context.



Ultralearners engage in direct practice. For example, instead of studying trivia from books or websites, *Jeopardy!* champion Roger Craig did simulated practices using past game shows.

- Go for project-based learning, not classroom-learning, e.g. learn computer coding by creating a computer game, or learn military history by writing a thesis about it.
- Use immersive learning. To learn a new language, immerse yourself in the culture (e.g. visit China to learn Mandarin).
- If you can't practice the skill directly, then use simulations, e.g. surgery or flight simulators.
- Aim for an "overkill", i.e. put yourself in a situation where the standards are much higher than your goal. If you want to become a great public speaker, don't just join the Toastmasters clubs. Take part in a public speaking contest.

KEY QUOTES

"The easiest way to learn directly is to simply spend a lot of time doing the thing you want to become good at."

"When we learn new things... we should always strive to tie them directly to the contexts we want to use them in."



4. Drill: Attack Your Weakest Point Relentlessly



In any process, a bottleneck can slow everything down, e.g. a limited vocabulary can hinder your fluency in a language. This principle is about isolating your key constraint, so you can focus on improving the 1 thing that'll enhance *all* other aspects of your skill.



To master a challenging skill, the best way is to break it down into small parts, master each component, then reassemble them. Ultralearners typically use the “Direct-Then-Drill” strategy:

- Start by practicing the skill directly in a situation that most closely matches the context intended for future use.
- Then, isolate the components of the direct skill that are limiting your overall performance or pace of learning.
- Ruthlessly drill those components before returning to your direct practice, to transfer the learning and see if the drills worked.



To design effective drills:

- Start by forming a hypothesis about what's holding you back, then use the Direct-Then-Drill approach to test if you're right.
- Use these tips for designing your drills:
 - (i) Use *time slicing* to isolate 1 action in a series of actions, e.g. practice the hardest few bars of a piece of music before playing the whole song.
 - (ii) Drill *cognitive components* e.g. pronounce pairs of words using different tones and record yourself speaking them.
 - (iii) *Copy the aspects of the skill you don't need*, so you can drill the aspects that you need. To present his arguments more logically, Benjamin Franklin rewrote the arguments from magazine articles from memory, then compared his version with the original to find his flaws. By “borrowing” others' arguments, he could focus on his writing.

KEY QUOTES

“Drilling problems without context is mind-numbing. However, once you've identified that it's the bottleneck preventing you from going further, they become instilled with new purpose.”

(iv) Spend more time practicing 1 component (e.g. your headline or first paragraph) to *magnify* that aspect.

- Regularly remind yourself on your goals to motivate yourself to persist in your drills.



5. Retrieval: Use Self-Testing to Learn



Retrieval practice is about trying to recall as much information as possible from memory, without referring to your materials. You learn much better when you (i) push yourself to recall info instead of reviewing it passively, and (ii) test yourself before you feel confident. Retrieval practice is the most effective when you can look up the answers immediately for what you don't know.



Self-testing is so effective because (i) it's the most similar to actual testing, (ii) it gives you specific feedback about what you know and don't know, and (iii) it helps you to measure your progress.

- The method doesn't *feel* effective since you'll find yourself struggling. However, people who tested themselves recalled much more than those who did passive review or used concept maps. The gap increased even more after 10 days.
- Psychologist R.A. Bjork coined the term "desirable difficulty" to describe how learning deepens when the brain has to work harder. The challenge is to test yourself just far enough to stretch your memory, but not so far that you've forgotten everything.
- In fact, you can use forward-testing to enhance your results. Take the final exam before the class even begins. This gives you a perspective of what's important, and prepares your mind to detect relevant information during lessons. In short, retrieval practice can improve both past *and* future learning.
- Use direct practice to determine what and how much to memorize. Basically, the things that come up regularly are the things that you must know in your head. Those that're rarely used can be looked up as and when needed.

KEY QUOTES

"The research is clear: if you need to recall something later, you're best off practicing retrieving it."

"Retrieval works to enhance future learning, even when there is nothing to retrieve yet!"

"Being able to look things up is certainly an advantage, but without a certain amount of knowledge inside your head, it doesn't help you solve hard problems."



Here are some useful methods for retrieval practice:

- Use digital or paper flash cards to pair a cue with a response, e.g. memorizing definitions or foreign-language vocabulary.
- Use free recall to memorize key points or arguments. After a lecture or reading a chapter, take a piece of paper and write down everything you can remember.
- Make notes by rephrasing the content in question-and-answer format, e.g. “When was the Magna Carta signed? In 1215.”
- If you need to practice certain skills (not just remember the information), create challenges to be solved later on your own.
- Apply closed-book learning by drawing a concept map of what you’ve just learned, without referring to the texts.



6. Feedback: Don’t Dodge the Punches



Feedback can be really painful and uncomfortable, but don’t avoid constructive criticism. Learn to manage your ego, and filter out the useful inputs from the noise so you know what to focus on.

- Famous comedians like Chris Rock are known to drop in at the Comedy Celler unannounced, just to test out their new materials on a small crowd to see what works and what doesn’t. They understand that when it comes to learning, nothing beats immediate, accurate and intense feedback.
- However, feedback is only useful if it provides information that guides your future learning and results. Feedback can actually backfire (i) if it focuses on the person (e.g. “you’re so smart” or “you’re so stupid”), or (ii) if the recipient doesn’t receive it in the right spirit.



There are 3 main types of feedback:

KEY QUOTES

“Retrieval is not a sufficient tool to create genius, but it may be a necessary one.”

“Everybody has a plan until they get punched in the mouth.”
– Mike Tyson

“Receiving feedback isn’t always easy. If you process it as a message about your ego rather than your skills, it’s easy to let a punch become a knockout.”

KEY QUOTES

- Outcome feedback tells you how well you're doing (e.g. whether you pass or fail, get an "A" or "C" grade). Such feedback is easy to get, and can motivate you to keep making progress. However, it doesn't tell you what to fix or how to improve.
- Informational feedback tells you what you're doing wrong, but it may not tell you how to fix it. An audience's bland response tells the comedian that his joke isn't funny, but it doesn't tell him *how* to make it funnier. Having said that, so long as the feedback is immediate, you can test different solutions to see what works, e.g. a programmer can keep changing his code until the error message disappears.
- Corrective feedback is the best type of feedback, since it tells you what you're doing wrong *and* how to fix it. However, this usually requires access to a teacher or coach, and the feedback is only as good as the person's expertise.



Generally, it's the most effective to combine immediate feedback with delayed review. Use these tactics to get better feedback:

- Filter out the background noise that distracts you from what's important. If you're reviewing the quality of your blog articles, don't just look at the volume of visitors. Look at how many people scroll to the end of your article or click on your links.
- Seek lots of rapid, high-intensity feedback. This is the best way to filter out the true insights from the noise.
- Adjust your tasks till you get the sweet spot for difficulty, i.e. you can't predict if you'll succeed or fail. That's when feedback truly becomes valuable.
- Look for metafeedback, or feedback on the effectiveness of your learning strategy. Analyze your learning rate (i.e. how fast you're improving in an area), or run experiments to compare which approach helps you to learn faster.

"Ultralearning isn't simply about maximizing feedback but also knowing when to selectively ignore elements of it to extract the useful information."

"Extract the signal from the noise, so you know what to pay attention to and what to ignore."

"Sometimes the easiest way to improve feedback is simply to get a lot of it a lot more often."



7. Retention: Stop the Leakages



To excel in any area, you need to remember stuff. Yet, our brain forgets things quickly and consistently, like a leaky bucket. Understand why you forget things, so you can deliberately use counter-strategies to remember things for the short-term and long-term.



Psychologist Hermann Ebbinghaus discovered the forgetting curve, which describes the brain's decreased ability to retain information over time. Unless the knowledge is consciously and repeatedly reviewed, we'll start losing it within days or weeks. Subsequent studies identified at least 3 theories to explain why we consistently forget so much:

- Our memories decay over time, and details fade away quickly. Generally, the farther an event, the harder it is for us to recall the details (unless it's unusually meaningful or vivid).
- In the human mind, memories often overlap, so new knowledge can create interference. Specifically, *proactive interference* occurs when past data makes it harder to acquire new knowledge (e.g. when they conflict with each other). Retrospective interference occurs when the new information "overrides" the old one.
- Some memories still exist in our minds, but we can't retrieve them because we've forgotten the cues. That's why we may feel like there's something we should remember, but we just can't recall it.



So, how can we counter our natural tendency to forget? Ultralearners deploy at least 4 mechanisms to improve their short-term and long-term retention:

- Space out your learning over intervals. It's better to spend 1 hour over 10 days learning something, than to cram the learning into 10 hours at a go. You can (i) use a system or schedule to methodically revisit a topic at increasingly longer intervals (e.g. start with daily, then weekly, monthly and 6-monthly revisions), or (ii) schedule projects to refresh important skills.

KEY QUOTES

"To retain knowledge is ultimately to combat the inevitable human tendency to forget."

"Many memories we have aren't actually forgotten but simply inaccessible."

"Our minds are a leaky bucket; however, most of the holes are near the top, so the water that remains at the bottom leaks out more slowly."

KEY QUOTES

- Proceduralize skills by embedding them into a sequence of steps, just like how typists can type without looking at the keyboard once the letters' position are deeply ingrained in their minds.
- Overlearn by practicing something even when you can already do it perfectly. The extra practice may not improve your performance, but it'll help you to recall the skills in future.
- Use mnemonics to translate hard-to-remember content (e.g. a sequence of digits) into vivid images or maps to facilitate recall.



8. Intuition: Dig Deep, then Build Up



True masters don't just memorize things. Mastery comes from a deep understanding of each component, how they're inter-related, and what the big picture looks like. Seek to truly understand what you're learning: immerse yourself in a subject so thoroughly that you form mental models of how things really work, and use play/exploration to develop your intuition.



Don't just study hard. Use these 4 rules to develop intuition:

- Don't give up too soon on hard problems. When you're about to give up on a problem, force yourself to keep going for just *10 more minutes*. New ideas and solutions may simply appear. Even if they don't, the extra effort will help you to notice and remember how a solution is derived later.
- Prove things to understand them. You'll never fully understand something just by reading someone else's results. Push yourself to prove or replicate the concept, e.g. to prove that you understand how a bicycle works, draw out all the parts in detail (e.g. gear, pedals) and show how they're connected.
- It's hard to learn abstract ideas, so start with a concrete example. For example, visualize how you'd apply a broad principle to a specific situation, or look beyond the spelling of a word to consider its deeper meaning, the feelings it evokes, etc.

"Do not ask whether a statement is true until you know what it means."
– Errett Bishop

- Don't fool yourself into thinking you know more than the subject experts. After all, you don't know what you don't know. Be willing to ask the "dumb" questions that everyone is afraid to ask.



Young developed the "Feynman Technique" based on his extensive research on genius mathematician Richard Feynman:

- Use this method to improve your understanding of an idea:
 - (i) At the top of a piece of paper, write down the *concept or problem* that you're trying to understand.
 - (ii) *Explain the idea* as if you had to teach it to someone else. Specify (a) how you'd explain the concept to someone who has never heard of it before, or (b) what's the process you'd use to solve the problem (and why).
 - (iii) If you're stuck, review your materials for the answer.
- The goal is to force yourself to articulate your understanding, so what you know (and don't know) becomes explicit. You can use this to (i) learn a new concept, (ii) tackle problems you can't solve, and (iii) expand your intuition on a complex topic.



9. Experimentation: Explore Beyond your Comfort Zone



When you're learning a new skill, it's usually enough to follow someone who's more successful. As you become more proficient, there'll be fewer people who can teach you. You'll also start to encounter new challenges that your predecessors haven't solved. To become a true master, you have to unlearn old methods and assumptions, explore new possibilities, expand/combine different skills/techniques, and discover your strengths and signature style.

- Vincent van Gogh didn't start painting till he was 26, and everyone told him he wasn't cut out to be an artist. Van Gogh decided to educate himself. He rigorously studied books and other artists' works, experimented with various materials, methods and philosophies, sought

KEY QUOTES

"Psychologists theorize that the difference between grand masters and novices is...that they have built up huge libraries of mental representations."

"Since many of our understandings are never articulated, it's easy to think you understand something you don't."

"As mastery becomes a process of unlearning over accumulation, experimentation becomes synonymous with learning as you force yourself to go outside your comfort zone and try new things."

feedback, and eventually developed his unique style. Unfortunately, he died at 37 and became famous only after his death.



Push beyond your comfort zone with 3 types of experimentation:

- Test out different learning resources, e.g. books, online materials, studying your peers, or practicing in various settings. Apply the insights from 1 resource aggressively, evaluate the results, then try something else.
- Try out various techniques by diving into a sub-topic or an area of specialization. For example, if you're trying to master a language, you can zoom into literature, comics, or business discussions, all of which will expose you to different skills and nuances.
- Explore different styles, so you can become aware of the full range of existing styles, dissect why other masters stand out, and emulate or integrate those concepts to create your own style.



Ultimately, you'll need to integrate your experiments back into your ultralearning projects, to target the problems you're solving. Here's how you can start experimenting:

- Copy then create. To build a foundation, start by copying what others are doing. When you've developed enough skills and knowledge, you can start to create your own style and direction.
- Use split-testing: Compare 2 methods side-by-side to see which method works better and why. You can eventually create a hybrid style that combines the best features from different approaches.
- Introduce new constraints to force yourself to try new solutions.
- Combine 2 or more unrelated skills to develop a new superpower. Scott Adams combined his engineering background with his skills as a cartoonist to create the highly-acclaimed Dilbert series.
- Explore the extremes, where there's much greater potential for creative breakthroughs than sticking to the

KEY QUOTES

"Careful experimentation not only brings out your best potential, it also eliminates bad habits and superstitions by putting them to the test of realworld results."

middle. Van Gogh used thick applications of paint in bold colors, both of which were unconventional for his time. You can always return to moderation after exploring the extremes—the key is to play with the full spectrum of possibilities.



This principle ties together the other 8 principles above. Learning is by nature full of uncertainty and trial-and-errors. It requires a growth mindset, where you continually try new things, obtain feedback, and adjust your course, to keep moving forward on your learning journey.



Getting Started



Ultralearning projects aren't easy, but they are well worth the effort. Your first ultralearning project is crucial as it will set the foundation for your future projects. Use these steps to improve your chances of success:

- Do your research. Define the topic you want to learn, the rough scope and objective, and the main resources you will use. Research how others have learned this skill successfully, identify the direct practices you can adopt, and consider the drills and backup materials you may need.
- Schedule your time. Decide how much time you'll commit, which slots you'll reserve in your weekly schedule, and the duration for your project. Schedule these into your calendar in advance. If your project will last for ≥ 6 months, try doing a pilot week to test out the arrangements before committing to it.
- Don't wait for the perfect plan. Start executing your plan, then periodically review/refine your approach with these questions:
 - (i) Metalearning: Have I done my research and spent $\approx 10\%$ of my total time preparing for my project?
 - (ii) Focus: Am I fulfilling my committed learning hours? Am I truly focused during those periods?

KEY QUOTES

"Sticking to the middle and playing it safe...allows you to explore only a subset of the total possibilities of your work."

*"The beginning is always today."
—Mary Shelley*

"What matters is the intensity, initiative, and commitment to effective learning, not the particulars of your time-table."

KEY QUOTES

- (iii) Directness: Am I learning in the way I'll eventually use the skill or knowledge?
- (iv) Drill: Am I tackling my weakest points? Am I working on small, manageable parts of a complex problem?
- (v) Retrieval: How do I know what/how much I've learned? Am I testing myself?
- (vi) Feedback: Am I getting honest feedback and applying them?
- (vii) Retention: What am I doing to retain what I've learned?
- (viii) Intuition: Do I deeply understand what I've learned?
- (ix) Experimentation: Am I trying new things beyond my comfort zone?
- Not every project will be successful. Review your results at the end of the project to figure out what went well, and what can be improved in future.
- Decide what to do with the skill: You can choose to:
 - (i) *Maintain it*. Decide how often/when to practice the new skills, and find ways to integrate it into your life. Start with more regular reviews, then space it out with time.
 - (ii) *Relearn it*. Even though it's not ideal to forget a skill, it's still easier to relearn something than to learn it from scratch.
 - (iii) *Master it*: Dive deeper into the skill through regular practice, or follow up with another ultralearning project, possibly to focus on a sub-topic or to transfer the skill to a new domain.



Ultralearning is the most effective way to learn a complex skill quickly, and to enhance your overall learning ability. However, it's not meant to totally replace traditional education or good habits. You can apply it alongside other alternatives:

- Low-intensity habits are great for accumulative learning or adding skills/knowledge through repetitive action, such as adding vocabulary to a language you already know.

"You're the one in charge, and you're the one who's ultimately responsible for the results you generate."

"The goal of ultralearning is to expand the opportunities available to you, not narrow them. It is to create new avenues for learning and to push yourself to pursue them aggressively rather than timidly waiting by the sidelines."

- Formal/structured education can also be useful for acquiring formal credentials, and for gaining access to certain communities or apprenticeship opportunities.



Other Details in the Book to Look out For

The book includes many examples of Young's own ultralearning projects as well as insights from other ultralearners in various fields. The book ends with a detailed case study of how the 9 principles were used by László Polgár to nurture his daughters into famous chess prodigies. Young also shares ideas on how to foster ultralearning at home, school or the workplace, and includes an appendix on his ultralearning projects. For more details, please visit ScottHYoung.com.

KEY QUOTES